AUTOMOTIVE DIAGNOSTIC AND ACTIVITY MONITORING

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***Abstract*—The Automotive Diagnostic and Activity Monitoring (ADAM) System is an integrated solution designed to provide real-time insights into the health and behavior of four-wheeled vehicles. Acting as both a diagnostic tool and a forensic black box, ADAM harmonizes multiple sensing modules—including a Global Positioning System (GPS), a gyroscope and accelerometer (MPU6050), and a vibration sensor (SW-420)—to observe vehicular dynamics. At its core, the system interfaces with the vehicle's Electronic Control Unit (ECU) via the On-Board Diagnostics (OBD) port, enabling deep access to engine parameters and fault codes.Sensor data is continuously collected, securely stored on an SD card, and synchronized with a PostgreSQL database. This rich dataset fuels a suite of machine learning models tailored for four key objectives: (1) Engine Health Monitoring, (2) Fuel Economy Analysis, (3) Driver Behavior Profiling, and (4) Predictive Maintenance. Insights derived from these models are visually rendered through an intuitive mobile application that presents the information via dynamic graphs and charts.To further enhance user engagement, the system incorporates an interactive AI-powered chatbot capable of responding to user queries and guiding necessary actions. Together, these components form a cohesive ecosystem—bridging vehicle intelligence with user accessibility, and paving the way for smarter, safer, and more informed driving experiences.**

***Index Terms*—Raspberry pi 5, On Board Diagnostic(OBD), Global Positioning System(GPS), ECU(Electronic Control Unit)**

I. INTRODUCTION

Over the last couple of years, vehicle monitoring has been a necessary measure to make vehicles and drivers safe, cut down on emissions to tackle climate change , and provide optimal performance. The significance of protecting human life and the necessity to obtain genuine information at the time of accidents for forensic analysis reinforce the importance of efficient monitoring systems. To serve this purpose, we created a vehicle monitoring logger on the basis of On-Board Diagnostics (OBD-II) port. The OBD-II port is a 16-pin standardized connector found in every contemporary vehicle,developed in the United States Of America in 1996, by the Society of Automotive Engineers (SAE). It was imposed to enable regulation of vehicle emission so as to ensure Environmental Protection Agency(EPA) standards are met. It is a diagnostic port used to monitor real-time readings of vehicle sensors and the Diagnostic Trouble Codes (DTCs) from the Electronic Control Unit (ECU), making it a valuable resource for monitoring vehicle performance.Ecu is a the main controller of the vehicle where all controls and actions are done.

OBD II port supports communication protocols such as ISO15765-4 (CAN-BUS), ISO14230-4 (KWP2000), ISO 9141-2, SAE J1850 VPW, and SAE J1850 PWM. This study is centered on the ISO 15765-4 (CAN-BUS) protocol that has been required on vehicles manufactured from 1996. The message request is send in a hexadecimal to the protocols and it gets interpreted according to one of the five OBD II protocols and sends back a hexadecimal response.Obd II uses two types of codes to request ECU data that are Diagnostic Trouble Code (DTCs) and Parameter Identifiers (PIDs).DTC are used to diagnostic malfunctions in various subsystem of the vehicle and PIDs are used to measure real time parameters

II. LITERATURE REVIEW:

In Recent years, there has been advancements in design and development of On board Diagnostic and also there are machine learning algorithms for predicting engine performance , fuel consumption and anomaly detection. In this section, we review the literature on design and development of OBD boards and developed machine learning algorithms.Additionally, we identify gaps in recent literature

A. Design and Development of On-Boards Diagnostic circuits

The study [1] contributes to the design of On Board Diagnostic systems that provides real-time information about a vehicle, such as engine speed, coolant temperature, pressure, and Diagnostic Trouble Codes (DTCs).Data Collection of OBD Using a keaz128 microcontroller. This paper [2] designs an Diagnostic system that has various data collection sources like gps,imu,camera and OBD port and stores locally for every 100 ms. Using BeagleBoard-XM with TI OMAP3530 processor logs the data. This paper [3] designs of Diagnostic system that collect obd data using ELM327 microcontroller along with gps .The data is transmitted to remote server and stored in sql database and transmit data using Carambola2 WiFi module.

B. Review of Machine Learning Algorithms

This study [4] is a detailed survey of machine learning methods used in engine performance, control, and fault diagnosis. It emphasizes the employment of Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forests (RF), and neuro-fuzzy models such as ANFIS for improving fuel economy and emission control. The review also discusses the use of AI in engine control with a special focus on reinforcement learning (RL) and deep reinforcement learning (DRL) for enhanced adaptability, efficiency, and real-time decision-making in sophisticated engine operations.

C. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [**?**].

Algorithm 1 IoT Sensor Data Collection and Transmission

Initialize:
  Initialize GPS module
  Initialize accelerometer/gyroscope
  Initialize vibration sensor
  Initialize OBD-II port connection
  Create empty dataframe df
  Connect to server
  Create main CSV file if not exists
while not exit_condition **do**
  $timestamp \leftarrow$ current time
  Collect sensor data:
    $gps\_data \leftarrow$ read_from_NEO6M()
    $accel\_data, gyro\_data \leftarrow$ read_from_MPU6050()
    $vibration\_level \leftarrow$ read_from_SW420()
    $obd\_data \leftarrow$ read_from_OBDII()
  Organize data into record:
    $record \leftarrow \{timestamp, gps\_data, accel\_data, gyro\_data,$
    $vibration\_level, obd\_data\}$
  Append to dataframe:
    $df \leftarrow df \cup record$
  Transmit to server:
    transmit_data_to_server($record$)
  Append to CSV:
    append_to_csv($record$, main_csv_file)
  Check conditions:
    $exit\_condition \leftarrow$ check_exit_conditions()
  Wait for next sampling interval
end while
Cleanup:
  Close all sensor connections
  Close server connection
  Save final state

References

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